

Md. Hasibul Hossain

Dhaka, Bangladesh

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[GitHub](#) — [LinkedIn](#) — [Portfolio](#)

Education

University of Rajshahi

Rajshahi, Bangladesh

Bachelor of Science in Information and Communication Engineering

CGPA: 3.53/4.00

Expected Graduation: July 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Natural Language Processing, Database Systems, Computer Networks, Data Structure and Algorithm.

Research Interests

- Artificial Intelligence, Machine Learning
- Computer Vision, Medical Image Analysis
- Natural Language Processing
- Cybersecurity

Research Experience

Open-World Instance Segmentation with Incremental Learning

April 2026 – Present

Undergraduate Researcher

- Built a segmentation system capable of detecting and incrementally learning new classes without forgetting, using YOLOv8-Seg, PyTorch, replay memory, and knowledge distillation.
- Achieved 72% mAP on known classes while maintaining 64% mAP after incremental learning, demonstrating robustness against catastrophic forgetting; paper in preparation.

Data Augmentation Strategies for Skin Lesion Classification

November 2025 – April 2026

Undergraduate Researcher

- Evaluated multiple augmentation policies across CNN architectures to improve skin lesion classification accuracy and robustness.
- Used Grad-CAM to interpret model decisions and analyze how different augmentations influence learned representations.

Uncertainty-Guided Adaptive Augmentation for Robust Skin Lesion Classification

February 2026 – June 2026

Undergraduate Researcher

- Developed an adaptive augmentation framework that dynamically adjusts augmentation strength based on Monte Carlo Dropout uncertainty estimates.
- Improved expected calibration error (ECE) and robustness against distribution shifts on several skin lesion datasets.

Publications / Manuscripts

- **Hasibul Hossain.** “Data Augmentation Strategies for Skin Lesion Classification.” *Under Review*, 2026.
- **Hasibul Hossain.** “Uncertainty-Guided Adaptive Augmentation for Robust and Calibrated Skin Lesion Classification.” *Under Review*, 2026.
- **Hasibul Hossain.** “Open-World Instance Segmentation with Incremental Learning.” *In Preparation*, 2026.

Technical Skills

- **Languages:** Python, C, C++, Java, JavaScript, TypeScript
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, YOLOv8, YOLOv8-Seg, BERT, RoBERTa
- **Web & Mobile:** React, React Native, Node.js, Express.js
- **Databases:** MySQL, PostgreSQL, MongoDB, Firebase
- **Tools & Platforms:** Git, GitHub, Linux, Docker, VS Code, Google Colab, Expo, Cloudinary

Selected Research Skills

- Deep Learning, Medical Image Analysis
- Incremental Learning, Model Calibration
- Data Augmentation, Explainable AI (Grad-CAM)
- Computer Vision, Object Detection, Segmentation

Projects

Smart Job Portal

April 2026

Tech Stack: React, Node.js, MongoDB

- Built a full-stack job portal with authentication, job posting/search, application tracking, and recruiter dashboard.

Campus Event Management App

November 2025

Tech Stack: React Native, Node.js, MongoDB, Firebase

- Developed a mobile app for campus event creation, ticket booking, and management with real-time notifications and image uploads.

LifeLine: Blood Donation Management System

May 2026

Tech Stack: MERN (MongoDB, Express, React, Node.js), Socket.io

- Created a blood donation platform featuring fast donor search, emergency request handling, and real-time chat between donors and requesters.

Achievements & Certifications

- **Champion** – Project Showcasing in Web Technology, International Girls in ICT Day 2026.

Leadership & Activities

- **Class Representative** 2023 – 2025
Department of Information and Communication Engineering, University of Rajshahi.
- **Department Football Team Captain**
Department of Information and Communication Engineering, University of Rajshahi.
- **BNCC (Bangladesh National Cadet Corps)**
Participated in cadet activities focused on discipline, teamwork, leadership, and community service.

References

Dr. Dipankar Das

Professor

Dept. of Information & Communication Engineering

University of Rajshahi

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Md. Golam Rashed

Professor

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